Aliah University

End-Semester Examination (Autumn Semester) - 2022

(For 5th Semester B.Tech (CSE))

Paper Name: Database Management System | Full Marks: 80

Paper Code: CSEUGPC11 | Time: 3 hrs

Answer all questions from Group-A and any 7 from Group-B

Group-A

1. Select the best alternative [10*1]

i) A□B is a trivial functional dependency if

A) $A \supseteq BA \supseteq B$ B) $A \sqsubseteq BA \sqsubseteq B$ C) $A = A \cap BA = A \cap B$ D) $A \cap B = A \cup B$ $A \cap B = A \cup B$

ii) $A \cap BA \cap B$ is same as

A) (A-B)-B B) (A∪ B)-A C) A-(A-B) D) A∪ (B-A)

iii) Project Operator in RA is equivalent to _________ in SQL

A) Project B) Select C) Group By D) Where

iv) After group by we can project

A) Any Attribute B) on which it is grouped C) Aggregated result D) B & C

v) An example of an aggregate function in SQL is

A) group by B) min C) exists D) all

vi) Grant statement is an example of

A) DDL B) DML C) DCL D) DQL

vii) After successful execution of the last instruction of the transaction, the state is

A) Partially committed B) Committed C) Ready of abort D) Aborted

viii) Rollback ensures –

A) Atomicity B) Durability C) Consistency D) Isolation

ix) Second level indexing is

A) Sparse B) Dense C) Hash D) Any

x) Starvation is associated with

A) Deadlock prevention B) Deadlock detection C) Deadlock recovery D) All

Group-B

2. What are the advantages of using DBMS over file based systems. Describe the three-level architecture of DBMS. [5+5]

3. Prove that Amstrong's axioms IR IV, IR V and IRVI using the fundamental rules. Define different types of keys with suitable example. [5+5]

4. Considering the following features of a RESEARCH PROJECT and design a ERD to represent the cardinalities of the entities : [10]

- Some employees are researchers
- Every project has a leader investigator

- Every project must be funded by an agency
- A project may include several topics
- A topic could appear in several projects
- Researchers must produce report(s)
- Each employee must have a supervisor

5. Find the primary key of the table R(A,B,C,D,E,F,G,H), considering the functional dependency set F= { A-->CD, AB-->EG, C-->G, D-->EF, AC-->D, G-->FH }. Discuss the phases of query optimization. [5+5]

6. Define ACID properties. What is cascade less schedule? What is irrecoverable schedule? What is the significance of precedence graph? Define 2PL protocol. [5*2]

7. Considering the tables Emp (empid,name,sal,deptid) and Dept (deptid,dname,dloc), write Relational Algebra for the following queries: [5 * 2]

   i. Find the name of the persons who earn below the average salary of her department.
   ii. What is number of employee who works at 'Ground floor'?
   iii. Who earn the maximum salary in 'Sales' department?
   iv. Find name of the department with maximum employee strength.
   v. Find the name of the person who earn the highest salary.

deptid G AVG(sal) (Emp ⋈ Dept)

8. Explain conflict and view serialization schemes with the following schedule [5+5]

T1	T2
R(A)	
W(A)	
	R(A)
	W(A)
R(B)	
W(B)	
	R(B)
	R(B)

(dloc G COUNT (dloc='GF' (Emp ⋈ Dept)))

9. Explain time-stamp based locking protocoi. Describe preemptive and non-preemptive schemes of deadlock avoidance. [5+5]

10. Explain dense and sparse indexing. Discuss dynamic hashing technique. [5+5]